

Analysis of speech recognition module for a robotic car

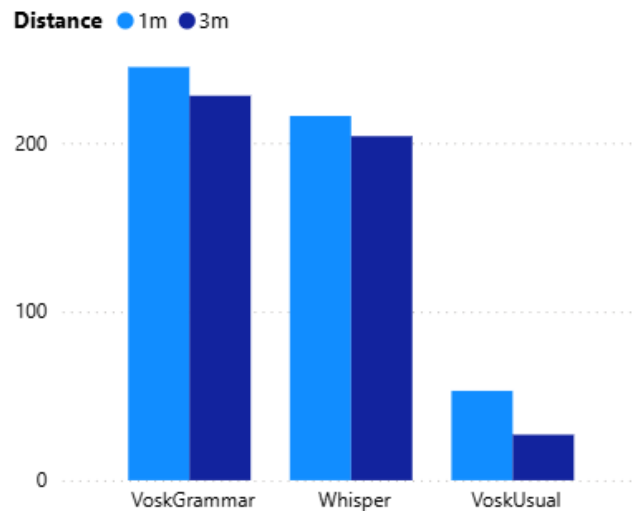
Rosmaster X3 Plus.

Based on my bachelor thesis.

Tested models: Vosk with grammar, Vosk without grammar and Whisper.

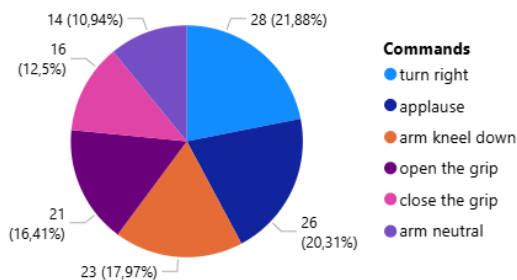
1. All models performed worse when the person was standing further from the microphone. This was expected, as speech quality naturally degrades with increasing distance.
2. Vosk with grammar showed consistently strong results at both distances. In contrast, the same model without a vocabulary achieved the lowest accuracy overall.

Speech recognition by distance

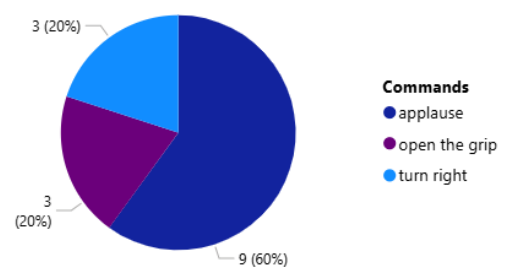


3. Vosk with grammar recognized all commands with approximately the same level of accuracy.
4. Whereas Vosk without a vocabulary failed to recognize some of the tested commands entirely.

Best recognized command

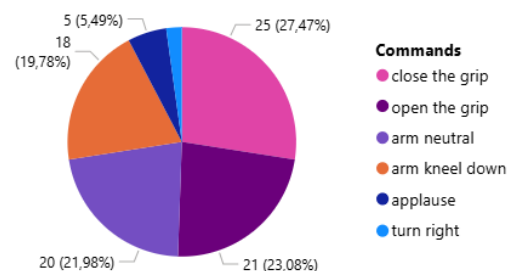


Best recognized command

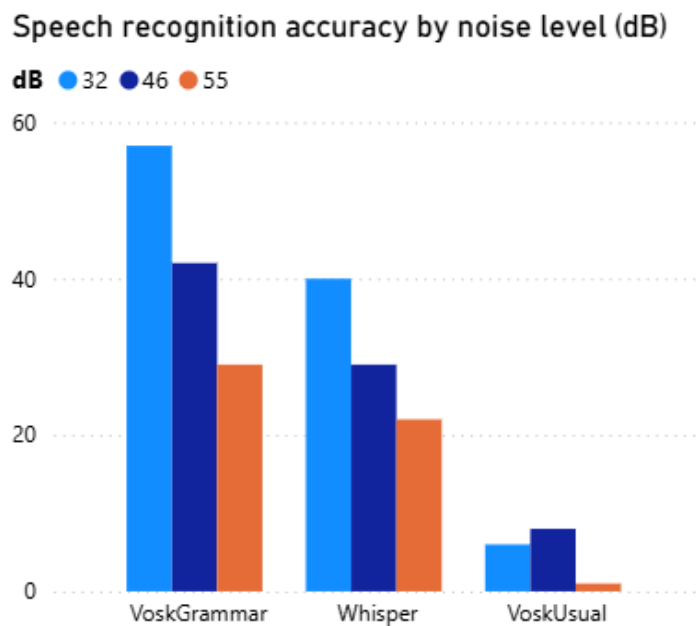


5. Whisper performed worse on short commands such as "applause" and "turn right", which may be related to the way the model was trained.

Best recognized command

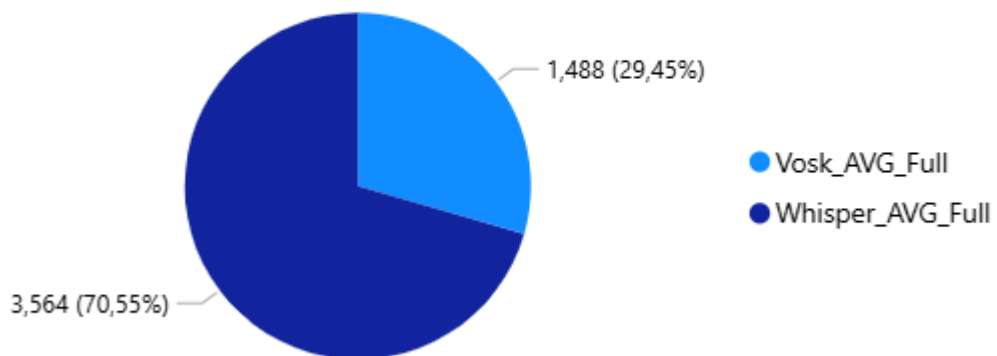


6. As expected, recognition accuracy decreases under louder noise conditions. However, Vosk without grammar achieved better accuracy at low noise levels, which may be related to stochastic resonance.



7. Vosk models recognized commands in an average of 1–2 seconds, while Whisper took 3–4 seconds.

Average command recognition time



Recommendations.

For the robot, I recommend using Vosk with grammar, as this model achieved the highest recognition accuracy and the fastest response time.